DAY 5: Stored Procedures and Triggers in MySQL

1. Stored Procedures in MySQL

A Stored Procedure is a precompiled collection of one or more SQL statements that is stored in the database and can be executed whenever required. It helps in reusability, better performance, and cleaner code.

Why Stored Procedures are used

- To reuse SQL logic multiple times
- To reduce network traffic
- To improve performance
- To maintain security by hiding logic
- To simplify complex queries

Example from SQL File

```
USE himesh;

DELIMITER $$
CREATE PROCEDURE tech()
BEGIN
   SELECT * FROM teachers;
END$$
DELIMITER ;

CALL tech();
```

Explanation of the Query

1. USE himesh; → Selects the database.
2. DELIMITER $$ → Changes delimiter so MySQL can understand multiple statements.
3. CREATE PROCEDURE tech() → Creates a procedure named tech.
4. BEGIN ... END → Contains SQL logic.
5. SELECT * FROM teachers; → Fetches all records from teachers table.
6. CALL tech(); → Executes the stored procedure.

2. Triggers in MySQL

A Trigger is a database object that automatically executes in response to specific events such as INSERT, UPDATE, or DELETE on a table. Triggers are mainly used to maintain data integrity and auditing.

Why Triggers are used

- Automatic data validation
- Auditing changes

- Enforcing business rules
- Maintaining data consistency

Key Points about Triggers

- Triggers execute automatically
- Cannot be manually called
- Associated with a specific table
- Use OLD and NEW keywords

Conclusion

Stored Procedures are useful for reusable logic and performance optimization, while Triggers are mainly used for automatic actions and maintaining data integrity. Both are important concepts for understanding database behavior.